Telco Customer Churn – Dataset / Project Proposal

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Project focus: Exploratory Data Analysis and preprocessing of the IBM Telco Customer Churn dataset for machine-learning readiness.

Project Purpose

The project examines customer information from a telecommunications company to identify patterns associated with customer churn. The dataset includes customer demographics, account information, services, contract type, payment method, monthly charges, total charges, and a Churn target. The goal of the project is to prepare a reproducible training dataset while preserving an untouched testing set for later model evaluation.

Research / Analysis Question

What customer characteristics and service/account patterns are associated with churn, and how can the dataset be cleaned and transformed so that it is suitable for machine-learning analysis?

Dataset and Method

The original IBM Telco Customer Churn sample dataset is documented on Kaggle. The workflow creates a 70/30 training/testing split using a fixed random state. EDA is performed only on the training data. Cleaning converts TotalCharges to numeric and handles its blank values; customerID is removed; binary variables are mapped to 0/1; multi-category variables are one-hot encoded; continuous variables are standardized; and RandomOverSampler balances the Churn classes in the training data. A NumberOfServices feature is also created from optional service indicators.

Expected Deliverables

Deliverable	Description
Final notebook	Integrated, documented .ipynb that runs from beginning to end.
Datasets	Training, testing, cleaned-training, and final processed-training CSV exports.
Visualizations	Matplotlib EDA and before/after preprocessing graphs.
Documentation	Kaggle source, final verification, and Person 5 contribution journal.
Repository	Organized files ready for the group's GitHub repository.

Dataset Source: Kaggle – https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Final integration: Jamie Lopez (Person 5)